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Effects of fuel efficiency improvements in personal transportation

Case of four-wheelers in India

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Abstract

Purpose – The purpose of this paper is to provide an assessment of the effects of fuel efficiency improvements on four-wheeler ownership, fuel consumption, fuel imports and emissions for personal transportation in the context of India. The paper also aims to measure the rebound effect induced by this policy.

Design/methodology/approach – The paper relies on a system dynamics model to analyse the problem. A causal loop model was developed initially, which was transformed to a stock and flow diagram. Simulation was carried out to capture the effects of fuel efficiency improvements in the four-wheeler sector of India.

Findings – The study has revealed that a policy of fuel efficiency improvements is favourable to the Indian four-wheeler sector growth, but this policy could result in an increase in fuel consumption and therefore a corresponding increase in the fuel imports and emissions in the country. This policy also induces direct rebound effect that adds up to the already alarming fuel consumption levels.

Research limitations/implications – The study is limited to four-wheeler passenger cars in India and gasoline as the transport fuel.

Originality/value – The paper offers a system dynamics model that can aid the government, vehicle manufacturers, and environment protection groups to further analyse policies regarding fuel efficiency improvements, fuel price adjustments, four-wheeler ownership, energy consumption and emission, and to obtain some useful policy insights before those policies are implemented.

Keywords India, Road vehicles, Gasoline, Fuel consumption, Fuel efficiency improvement, System dynamics, Four-wheeler ownership, Direct rebound effect, Emissions

Paper type Research paper

1. Introduction

Transportation fuel vulnerability is one of the issues currently faced by India. India is ranked the third most fuel vulnerable country among 26 other countries in the world (Gupta, 2008). The stagnant production in the existing oil fields and low extraction rate has slowed down the fuel supply pushing the country to be fuel import dependent (Choudhary and Shukla, 2006; Tonnesson and Kolas, 2006). At present, India meets about 80 percent of its fuel requirements through imports at international prices; major share of which is consumed by the Indian transport sector (Ghosh, 2005). High fuel imports made at international prices has resulted in frequent increase in fuel prices in the country (Ghosh, 2009; Subramanian *et al.*, 2005). Moreover, high fuel consumption rates have resulted in high greenhouse gas emission levels in the country. In light of

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increasing fuel consumption and price, the policy adopted recently by the Indian Government and vehicle manufacturers is to improve the technical fuel efficiency of the vehicles (Anand, 2008; Ritterspach and Ritterspach, 2009).

The fuel efficiency improvement is a strategy that is adopted all over the world to overcome the vulnerability of supply and surging prices, and to mitigate the growing greenhouse gas emissions (Barla *et al.*, 2009). Existing studies on the effectiveness of this policy in other parts of the world show that the policy was not wholly successful in reaping the expected benefits out of it (Sorrell and Dimitropoulos, 2008). India has also adopted this policy with an aim to reduce its consumption of fuel, imports and the resulting emissions.

In this paper, we analyse the effects of fuel efficiency improvements on four-wheeler ownership, fuel consumption, fuel imports and emission in India using system dynamics tools. The study also captures and quantifies the direct rebound effect which is a consequence of the fuel efficiency improvements. A causal model is proposed in the paper to capture the dynamic interaction between variables in the system considered which includes the four-wheeler stock, four-wheeler trips and distance traveled, fuel price, fuel consumption, fuel imports and emission. This causal loop model is further transformed to a dynamic stock and flow diagram to simulate the effects of fuel efficiency improvements in the Indian four-wheeler sector.

The rest of the article is organized as follows. Section 2 is on the fuel efficiency improvement policy adopted worldwide and in India. Section 3 is on direct rebound effect. Section 4 describes the system dynamics methodology and gives a description of energy modeling with system dynamics. Section 5 presents the causal model developed for the system considered in the Indian context. Section 6 gives the dynamics of the fuel efficiency improvements with the help of simulation results. Finally the conclusions are laid out.

2. Fuel efficiency improvement policy – worldwide and in India

Surging oil prices and growing concerns over climate change have renewed policymaker interest in raising fuel efficiency standards (Barla *et al.*, 2009).

2.1 Fuel efficiency improvement policy adopted worldwide

As a part of energy conservation and emission abatement initiative, energy efficiency improvements are being attempted in all energy intensive sectors around the globe. Such improvements are also being necessitated in order to fulfill the need to reduce the greenhouse gas emissions that leads to serious environmental problems. The US Congress recently has decided to increase the fuel efficiency of new four-wheelers by ten miles per gallon over the next ten years. Japan is implementing regulations requiring automakers to improve fuel efficiency by 20 percent by 2015. The European Commission is to impose a standard of limiting carbon dioxide emissions to 120 or 130 grams per kilometer by 2012. The Canadian Government has announced the introduction of new mandatory standards on fuel efficiency improvements over the existing (Barla *et al.*, 2009). Moreover, manufacturers of energy consuming products have adopted the policy of energy efficiency improvements with the aim of reducing the energy consumption and the associated social impacts. Perhaps every major automobile manufacturing economy in the world has adopted some form of fuel economy standards for motor vehicles (An *et al.*, 2007). According to the "world energy outlook" of International Energy

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Agency (IEA, 2009), carbon abatement that would limit the global rise of temperature to 2°C could be possible only through improvements in energy efficiency.

2.2 Fuel efficiency improvement policy in India

The transport sector is the major consumer of fuel in India (Ghosh, 2005) and high consumption has made the country highly dependent on oil imports (Ghosh, 2009; Subramanian *et al.*, 2005). India is slowly turning into a country which is almost entirely relying on the import of transport fuel. High fuel imports, made at international prices, have resulted in frequent increases in fuel prices in the country. Indian retail price for gasoline is among the highest in South-Asia and other developing Asian regions; even exceeding prices of more developed economies (IEA/OECD, 2006). Thus, surging international oil prices, at which imports are made, influence the fuel price rises in India which in turn have an impact on the sales in four-wheeler segment (Anand, 2008). Owing to this fuel price rises, the Indian customers pro-actively demand for fuel-efficient vehicles (Marathe, 2006), which motivates the vehicle manufacturers to improve the fuel efficiency of their vehicles.

Pollution from the automobile industry has significant implications on the environmental condition in India. The need to mitigate environmental pollution from the transport sector has led to the adoption of emission control practices in 1991 through fuel efficiency regulations in the country. Although India's emission control norms are already aligned with the Euro standards (Bharat stage I, II, III and IV equivalent to Euro standards), the country lags behind the Euro norms by about seven to eight years as far as implementations are concerned. Steps are adopted since 1996 to improve the fuel quality in order to bring about vehicle fuel efficiency improvements in India. Recently, the maximum limit of benzene availability in petrol has been set at 5 percent for the country (3 percent for the metros); and the sale of leaded petrol has been discontinued throughout the country (SIAM, 2011). In June 2009, Indian Government announced that it would soon make fuel efficiency labeling of cars mandatory (BEE, 2009; Kojima, 2009).

Improvement of technical fuel efficiency is the policy adopted by the Indian Government and the four-wheeler manufacturers to overcome the fuel crisis (crisis of high consumption and low availability), to sustain in the market in the face of rising fuel prices (Anand, 2008; Ritterspach and Ritterspach, 2009), and to fight the increasing emissions. Taking into consideration the importance of improving the fuel efficiency of vehicles in India in mitigating fuel consumption, imports and emissions, the Automotive Research Association of India (ARAI) has initiated further research in this line (Marathe, 2006). Need exists, therefore, to study the effects of the fuel efficiency improvement policy on the four-wheeler ownership, fuel consumption, fuel imports, and emission levels in the country.

3. Direct rebound effects

The energy efficiency improvement efforts that aim for conservation of energy and reduction of energy imports and harmful emissions, however, also give rise to the much discussed "Rebound Effect". Although a vast pool of literature exists that enumerates the concept, and discusses its measurement and quantification methods (Herring and Sorrell, 2009; Sorrell and Dimitropoulos, 2008; Sorrell *et al.*, 2009; Sorrell, 2009); the literature is still sparse that discusses the rebound effect in the context of India.

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The rebound effect occurs when an increase in the energy efficiency of a particular service decreases the overall cost of that service, resulting in an increase in usage and hence an increase in energy consumption (Barker *et al.*, 2009; Berkhout *et al.*, 2000; Greene, 2010). This is known as direct rebound effect. There are two more types of rebound effects namely indirect and economy wide rebound effects. For the present study only direct rebound effect is considered.

The "energy service cost" (or energy service price) is a general term used in energy literature and in case of transportation energy, it implies the fuel cost per vehicle-kilometer (Nassen and Holmberg, 2009). The fuel cost is the chief variable monetary cost of vehicle travel (i.e. travel cost) to which fuel price is the main "contributor" for a vehicle user (Greene, 2010). Travel cost (TC) per kilometer is dependent on fuel efficiency (FE) (in terms of miles/gallon) as well as fuel price (FP) per gallon and is given by the identity TC = FP/FE. This identity of travel cost shows that an improvement in fuel efficiency reduces the travel cost for a vehicle user. However, reduced travel cost (TC) encourages the vehicle users to travel more. Thus, the "vehicle miles traveled" (VMT or distance traveled using the vehicle by user) increases, and so is the fuel consumption (FC) since FC = VMT/FE. Thus, the effects of fuel efficiency improvements are offset by a corresponding (or sometimes more) increase in travel. Additionally, reduced travel cost encourages more sales of vehicles which increase the vehicle ownership levels leading to a further increase in the fuel consumption. This phenomenon is known as the "direct rebound effect" (Roy, 2000; Small and Van Dender, 2007; Stepp *et al.*, 2009).

The magnitude of the rebound effect for motor vehicles is of interest from a policy perspective because it may generate external costs of traffic congestion and tailpipe emissions that have an impact on the society (Greene, 2010). The direct rebound effect for personal transportation is measured using the travel cost (P) elasticity and vehicle distance traveled (VMT) utilising the identity below:

$$\eta_{\mu}(E) = -\eta_{P}(S) - 1$$

The above identity measures the direct rebound effect by obtaining the relationship between energy efficiency (μ) and energy consumption (E) through the price elasticity of the demand for useful work ($\eta_P(S)$); where the useful work (S) for transport sector is the vehicle miles driven or VMT (Berkhout *et al.*, 2000; Greene, 2010; Sorrell and Dimitropoulos, 2008). For the present study, we have utilised this identity for measuring the direct rebound effect in personal transport sector in India.

4. System dynamics and energy modeling

System dynamics is a method of analysing system related problems in which time is an important factor and involves the study of how a system can be defended against or benefited from the shocks that falls upon it from the external world (Coyle, 1977). It is rooted in feedback control theory and nonlinear dynamics. System dynamics is used to capture the feedback structure of the system which gives rise to the behavior of the system over time and aids in carrying out policy experimentation.

System dynamics is widely applied in energy modeling. System dynamics have been used to study the market penetration of alternative fuel vehicles (Gillingham and Leaver, 2008; Janssen *et al.*, 2006; Struben and Sterman, 2007; Supple, 2007), to model the urban transportation related issues of road congestion, modal-split (Khanna *et al.*, 1985; Wang *et al.*, 2008) and emissions (Piattelli *et al.*, 2002; Stepp *et al.*, 2009). Kiani *et al.* (2010)

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gives a detailed survey on the application of system dynamics in energy system modeling. Meadows *et al.* (2004) has utilised system dynamics in the study of nonrenewable energy resources. System dynamics have been utilised to study the discovery life cycle of natural gas (Naill, 1972) and for estimating petroleum resources (Davidsen *et al.*, 1990; Sterman *et al.*, 1988; Sterman and Richardson, 1985). Qudrat-Ullah and Karakul (2007) have used system dynamics to study the electricity supply sector in Pakistan in the context of environmental and resource constraints. Moreover, Bhattacharyya and Timilsina (2010) point out that system dynamics was one of the six energy model categories used by Hoffman and Wood (1976) to categorise the energy models. This vast literature shows the suitability of system dynamics in transportation and energy systems modeling.

A causal loop diagram (causal model or causal loop model or CLD) forms the initial step in system dynamics modeling. CLDs show the causal relations between the system entities through the use of text, arrows and symbols (Stepp *et al.*, 2009). An arrow having its tail at the cause variable and its head at the affected variable represents the relations between the same. The arrow head bears the causal link polarity. A positive polarity indicates that the perturbation in the causal variable is transmitted in the same direction to the effect variable and a negative polarity indicates that the perturbation in the causal variable is transmitted in the opposite direction to the effect variable provided *ceteris paribus*. The causal relations finally form feedback loops that are either reinforcing (R) or balancing (B) in nature (Sterman, 2000). The causal loop diagram is the input to the final quantitative modeling involving flow diagrams that will bring out the behavior over time of the system being studied.

Causal models are developed heuristically involving iterations which reconsider the model boundary in order to include the feedback connections appropriate for a study. The adjacency matrix method (Ali and Ramaswamy, 1993; Camara, 1991) can be used for this purpose. The available literature is studied initially and is decomposed into a sequence of inferences. Nouns and adjective/noun forms in every inference are identified and integrated into a matrix to facilitate selection of variables and polarized relationships. This adjacency matrix is finally translated into the causal loop diagram.

5. Developing a causal model for fuel efficiency improvement in Indian context

In this section, a causal loop model is presented to capture the effect of fuel efficiency improvement policy in Indian context. The model considers the effect of this policy on four-wheeler ownership, four-wheeler distance traveled, fuel consumption, fuel imports and emission in India. The model also captures the direct rebound effect induced by this policy.

An overview diagram of the system is shown in Figure 1. This overview diagram defines the boundary of the system considered in the present study. The causal model developed is shown in Figure 2. Wherever the system variable is repeated in the causal model, the same is duplicated within brackets (< >). The delays in the system is presented by two parallel lines crossing the causal links (//). In the present causal model, only the main loops are labeled to avoid confusion in interpreting the model. Table I gives the detailed description of the variables used in the causal model. Table II gives the details on main labeled loops, loop polarity, loop name and loop components.

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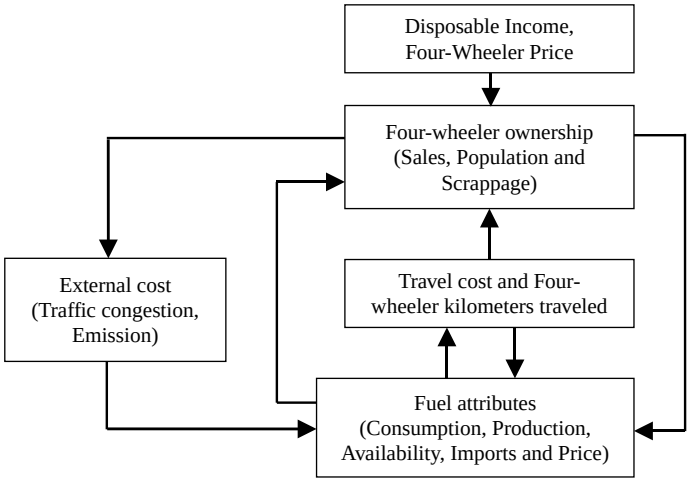


Figure 1.
Overview diagram for the developed causal model

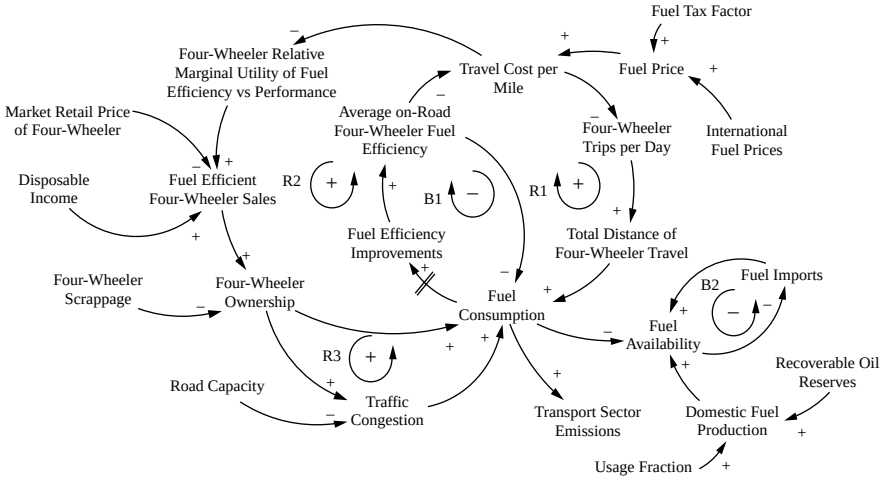


Figure 2.
Complete causal model for the whole system

The causal model shown in Figure 2 captures the effect of fuel efficiency improvement policy on four-wheeler ownership, fuel consumption, fuel imports and emission, the understanding of which are important in making transport and energy related policies of mitigating the effects of fuel price rises, increasing the four-wheeler ownership levels, reducing the fuel consumption, fuel imports, traffic congestion and emission in India.

5.1 Effect of fuel efficiency improvements on four-wheeler ownership

Fuel efficiency is a very important consideration for a consumer who intends to own a four-wheeler. Studies reveal that consumers in India are increasingly demanding more fuel efficient vehicles (Anand, 2008; Marathe, 2006). Surging fuel prices is one of the most important reasons that trigger the adoption of fuel efficiency improvement policies (Barla *et al.*, 2009).

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Variable	Description	Units/graph function	Endogenous/ exogenous	Loop to which the variable belongs
Fuel price	The price of a gallon gasoline in Indian markets	Dollars/gallon	Endogenous	No
Fuel efficiency improvements	The fuel efficiency of four-wheeler population claimed by manufacturers	Miles/gallon of fuel	Endogenous	R1, R2, R3, B1
Average on-road four-wheeler fuel efficiency	Distance traversed per gallon of fuel consumed	Miles/gallon of fuel	Endogenous	R1, R2, R3, B1
Domestic fuel production	Indigenous production of fuel in India	Gallons/year	Endogenous	No
Disposable income	Income of Indian urban class after payment of taxes	Dollars/year	Exogenous	No
Market retail price of four-wheeler	Price for a four-wheeler on-road in India	Dollars	Exogenous	No
Four-wheeler relative marginal utility of fuel efficiency vs performance	The ratio of consumer utility of one mile per gallon of fuel efficiency to one unit of performance	Units of utility / miles per gallon	Endogenous	R2, R3
Four-wheeler ownership	Total four-wheeler population in India for a period of one year	Four-wheelers/ year	Endogenous	R2, R3
Fuel efficient four-wheeler sales	Total fuel efficient four-wheeler sales in India for a period of one year	Four-wheelers/ year	Endogenous	R2, R3
Four-wheeler scrappage	The number of four-wheelers scrapped each year	Four-wheelers/ year	Exogenous	No
Four-wheeler trips per day	Total number of trips made by a four-wheeler user per day	Number of four-wheeler trips/ day	Endogenous	R1
Fuel consumption	Amount of fuel consumed per year by the vehicle population in India	Gallons/year	Endogenous	R1, R2, R3, B1
Travel cost per mile	The cost incurred to the consumer per four-wheeler mile driven also known as energy service cost	Dollars/mile	Endogenous	R1, R2, R3
International fuel prices	Fuel price at which India imports transport fuel from international markets	Dollars/barrel	Exogenous	No

(*continued*)

Table I.
Description of the variables used in the causal model

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Variable	Description	Units/graph function	Endogenous/ exogenous	Loop to which the variable belongs
Fuel tax factor	Fuel tax imposed on fuel sold in India by the government expressed as a fraction	Dimensionless	Exogenous	No
Fuel availability	Availability of gasoline for use in India	Gallons	Endogenous	B2
Fuel imports	Total quantity of fuel imported by the Indian Government each year	Gallons/year	Endogenous	B2
Recoverable oil reserves	The oil reserves in India that is recoverable	Gallons	Exogenous	No
Usage fraction	The fraction of recoverable oil reserves produced in India	Percentage	Exogenous	No
Total distance of four-wheeler travel	The total distance traveled each year by the four-wheeler population	Miles/year	Endogenous	R1
Traffic congestion	It is the total number of vehicles on the road at a given time expressed as a ratio of vehicle volume to road capacity	Dimensionless	Endogenous	R3
Road capacity	The capacity of Indian roads to hold the maximum number of vehicles	Passenger car units	Exogenous	No
Transport sector emissions	Annual emissions due to four-wheeler population in India	Metric tons/ year	Endogenous	No

Table I.

Fuel efficiency and performance are considered as important attributes of consumer preference (Stepp *et al.*, 2009; Turrentine and Kurani, 2007). In the context of rising fuel prices in India, the consumers highly prefer the fuel efficient four-wheelers for which they are willing to pay and which is often known as "stated preference" in four-wheeler ownership modeling (Brownstone *et al.*, 2000; Kumar and Rao, 2006; Ritterspach and Ritterspach, 2009). A fuel efficient four-wheeler consumes less amount of fuel for the same distance traveled which means a reduction in travel cost incurred to the user, and which is known as "first-order effect" (Berkhout *et al.*, 2000). This attracts the consumers towards fuel efficient four-wheelers which will increase its sales and ownership, and boost the growth of this sector. Hence, the fuel efficiency improvement policy favours the four-wheeler sector growth in the country. But an increase in four-wheeler ownership will result in an increase in fuel consumption levels and related

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Loop ID	Balancing (−) or reinforcing (+) loop	Loop name	Loop components
R1	Reinforcing	Rebound effect loop	Fuel efficiency improvements, average on-road four-wheeler fuel efficiency, travel cost per mile, four-wheeler trips per day, total distance of four-wheeler travel, fuel consumption
R2	Reinforcing	Fuel efficiency – four-wheeler ownership loop	Fuel efficiency improvements, average on-road four-wheeler fuel efficiency, travel cost per mile, four-wheeler relative marginal utility of fuel efficiency vs performance, fuel efficient four-wheeler sales, four-wheeler ownership, fuel consumption
R3	Reinforcing	Four-wheeler ownership – traffic congestion loop	Fuel efficiency improvements, average on-road four-wheeler fuel efficiency, travel cost per mile, four-wheeler relative marginal utility of fuel efficiency vs performance, fuel efficient four-wheeler sales, four-wheeler ownership, traffic congestion, fuel consumption
B1	Balancing	Fuel efficiency – fuel consumption loop	Fuel efficiency improvements, average on-road four-wheeler fuel efficiency, fuel consumption
B2	Balancing	Fuel vulnerability loop	Fuel imports, fuel availability

Table II.
Details of causal model on feedback loops, loop polarity, loop name and loop components

fuel imports and emissions in the country. This in turn calls for a further improvement of four-wheeler fuel efficiency as evidenced from loop R2 (Figure 2). The four-wheeler sale is also influenced by the price and income of the people. Hence, these variables are considered as exogenous variables in the present study.

5.2 Effect of fuel efficiency improvements on rebound effect

Fuel efficiency improvements favor the four-wheeler users through a reduction in travel cost by reduced fuel consumption. This is known as "price effect" in energy efficiency literature (Nassen and Holmberg, 2009). This reduction in travel cost per unit distance encourages the users to have more four-wheeler trips per day. Thus, an increase is observed in the vehicle miles traveled (VMT). This behavior finally results in an increase in fuel consumption thereby demanding a further improvement in vehicle fuel efficiency. Thus, a rebound effect has occurred due to the so-called "price effect" (loop R1 in Figure 2). The increase in fuel consumption due to this direct rebound effect leads to an increase in the fuel imports and emissions in the country.

5.3 Effect of fuel price on energy service cost

The driving habits of four-wheeler users are influenced by the fluctuations in fuel prices. When there is an increase in fuel prices, the cost of driving increases (evident from the identity TC = FP/FE from Section 3) which encourages users to drive less. This reduces the trips per day and hence the distance traveled by the four-wheeler that considerably reduces the transport sector fuel consumption, fuel imports and emission in the country. Thus, the fuel price rises can act as a control over the increased fuel consumption due to rebound effect as observed by Greening *et al.* (2000) and Roy (2000). But to what extent this fuel price increase is favourable requires further studies.

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5.4 Effect of fuel efficiency improvements on traffic congestion

Fuel efficiency improvement policy can have an impact on traffic congestion on-roads due to an increase in four-wheeler ownership levels (Greene, 2010; Greening *et al.*, 2000). Hence, changes in four-wheeler ownership levels in response to the improvements in fuel efficiency of the four-wheelers have an impact on external cost of traffic congestion which in turn considerably influences the transport sectors' contribution to fuel consumption and emission in India.

The proposed causal model considers road traffic congestion as an external cost of increased four-wheeler ownership levels. The fuel efficiency improvement policy in India can increase the four-wheeler ownership levels resulting in an increase in road traffic congestions in India (loop R3). Increased congestion in turn increases the fuel consumption rates by the four-wheelers. Increased fuel consumption implies more fuel imports and emissions in the country. Hence, an increase in four-wheeler ownership and traffic congestion leading to increased fuel consumption as a result of fuel efficiency improvements have a positive influence on the transport sector emissions which is considered in the causal model.

6. Dynamics of fuel efficiency improvements

In order to obtain the dynamics of fuel efficiency improvements, a stock and flow diagram was developed using STELLA 5.0 software. The stock and flow diagram has considered all the causal relationships between all the exogenous and endogenous variables considered in the study (the Appendix). The model is based on the data collected from reliable Indian Government approved data sources like Centre for Monitoring Indian Economy (CMIE), Business Beacon (BB), Central Statistical Organisation (CSO), Oil and Natural Gas Corporation of India (ONGC), and Society of Indian Automobile Manufacturers (SIAM). A dimensional analysis was carried out to ensure the dimensional consistency of the model (Wolstenhome, 1990). Model simulation starts from the base year 2010 and is for a time horizon of 12 years.

The behavior of the model is highly dependent on the values considered for the gross domestic product (GDP) and the fuel price in the Indian context for the next 12 years. It is assumed that while the GDP will follow an 8 percent growth rate per year as it is presently, the fuel price in India will continue to rise and will reach Rs 180 per liter at the end of 12 years.

The simulation is carried out for two scenarios. Scenario-1 is the base run with no fuel efficiency improvements. Scenario-2, on the other hand, considers the on-going fuel efficiency improvement efforts in the country. The dynamics of fuel efficiency improvements are discussed in the sections that follow.

6.1 Scenario-1: base run with no fuel efficiency improvements

Initially the system dynamics model was run with no fuel efficiency improvements undertaken as the scenario-1. The simulation results for important variables in scenario-1 are summarized in Table III and are graphically shown in Figures 3 and 4. From Figure 3, it is evident that the four-wheeler ownership is growing in the country resulting in increased traffic congestions. This increase in four-wheeler ownership is due to an increase in disposable income of the Indian urban class. Owing to the increasing fuel prices and no improvements in fuel efficiency, the vehicle distance traveled (VMT) is registering a downward trend indicating a demotivation

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Year	Fuel efficiency improvements (percent improvement)		Four-wheeler ownership (millions)		Fuel consumption (metric tons)		Fuel imports (metric tons)		Emissions (metric tons)	
	S-1	S-2	S-1	S-2	S-1	S-2	S-1	S-2	S-1	S-2
2010	1	1	20.24	20.24	13,072	13,072	9,472	9,472	114,638	114,638
2012	1	6	25.08	25.28	15,546	16,665	11,586	12,705	136,337	146,156
2014	1	11	29.44	30.15	17,501	19,867	13,181	15,547	153,480	174,237
2016	1	15	33.41	34.79	19,070	22,749	14,390	18,069	167,242	199,510
2018	1	18	37.02	39.21	20,160	25,130	15,120	20,090	176,806	220,391
2020	1	22	40.33	43.43	21,013	27,318	15,613	21,918	184,282	239,579
2022	1	25	43.38	47.47	21,644	29,234	15,878	23,438	189,817	256,384

Notes: S-1 corresponds to scenario-1: no fuel efficiency improvements; S-2 corresponds to scenario-2: on-going fuel efficiency improvements

Table III.
System dynamics simulation results for scenario-1 and scenario-2

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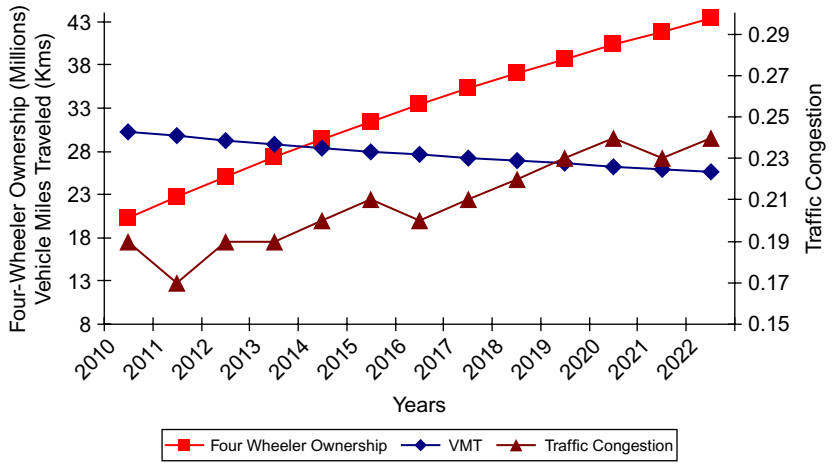


Figure 3.
Four-wheeler ownership and traffic congestion for scenario-1

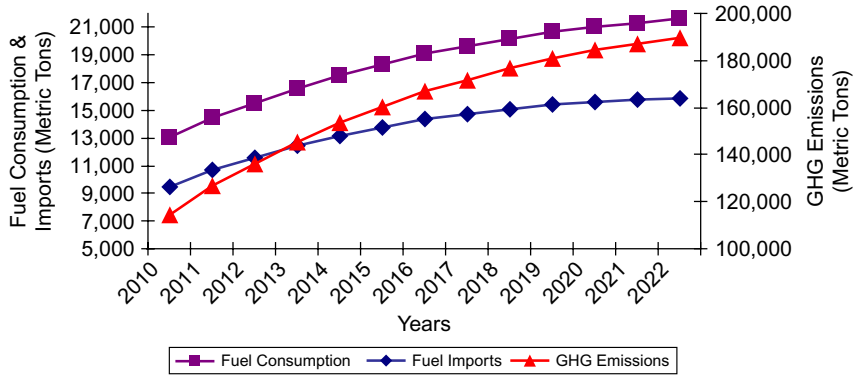


Figure 4.
Fuel consumption, fuel imports and emissions for scenario-1

of the four-wheeler users in driving more. However, increase in the ownership more than offsets the fall in VMT, and as a result, fuel consumption, fuel imports and emissions show an increasing trend for the next 12 years (Figure 4).

6.2 Scenario-2: with on-going fuel efficiency improvements

The fuel efficiency improvement policy affects the four-wheeler sales in India. From consumers' side, owing to the increase in fuel prices, consumers are attracted towards fuel efficient four-wheelers as it reduces the travel cost. In the present model, we have considered the four-wheeler sales to be determined by the four-wheeler price, disposable income and the four-wheeler marginal utility of fuel efficiency to performance. The disposable income is considered for the urban people as they are found to be the potential customers for four-wheelers in the country. The four-wheeler scrappage is determined by the average age of the four-wheelers which is taken to be 15 years. The European Union (EU) fuel efficiency standard, followed by India, is adopted from An *et al.* (2007), and it is assumed that the same trend will follow in India from the year 2010 onwards. The EU fuel efficiency standard shows 16 percent improvement in vehicle fuel efficiency

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in a period of six years and 23 percent improvement in a period of 11 years. These values are shown in the stock and flow model by the variable "improved fuel efficiency" (see the simulation results shown in Figure 5 and Table III).

A coflow modeling approach is adopted to model the four-wheeler ownership and to capture the on-road fuel efficiency of the four-wheelers (the Appendix). The on-road fuel efficiency is the realised fuel efficiency for four-wheelers on Indian roads. Coflow models are recommended modeling approach for any aging chains as it is for the use and scrappage of cars (Sterman, 2000). These models are also utilised in tracking the fuel efficiency of different vintages of cars (Vensim, 2003). The coflow model consists of two simultaneous flows – one of four-wheeler ownership and the other of four-wheeler ownership multiplied by their respective fuel-efficiencies. The average on-road fuel efficiency will be obtained by dividing the latter by the four-wheeler ownership.

Figure 5 shows the simulation results of the model with regard to actual improved fuel efficiency, realised on-road fuel efficiency and the four-wheeler ownership in the country.

The simulation result shows that as the fuel efficiency improvements (i.e. average on-road fuel efficiency) takes place the four-wheeler ownership also increases in the country. This is because of the positive influence of the fuel efficiency improvements on the four-wheeler demand and sales in the Indian market. However, this increase in four-wheeler ownership has further impacts on the fuel consumption, fuel imports and emission in the country.

Additionally, the simulation results show that with the base year of 2010, the present trend of fuel efficiency improvements continuing in future will lead to 70 percent of the total four-wheelers stock of India to be fuel-efficient by 2022.

6.2.1 Effect of fuel efficiency improvements on vehicle distance traveled (VMT) and traffic congestion. Improvements in fuel efficiency reduce the travel cost as the four-wheeler consumes less amount of fuel owing to these efficiency improvements. This in turn motivates the four-wheeler user to drive more that results in an increase in four-wheeler distance traveled (VMT) which is evident from the system dynamics simulation results shown in Figure 6. Increase in four-wheeler distance traveled will surely increase the fuel consumption, fuel imports and emission levels in the country.

Fuel efficiency improvement policy can result in social impact of traffic congestion as explained in Section 5.4. The traffic congestion is usually measured as a dimensionless

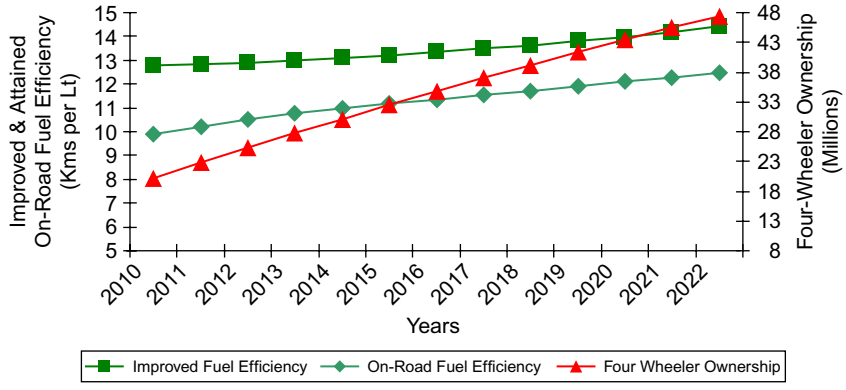


Figure 5.
Four-wheeler ownership growth in response to fuel efficiency improvements

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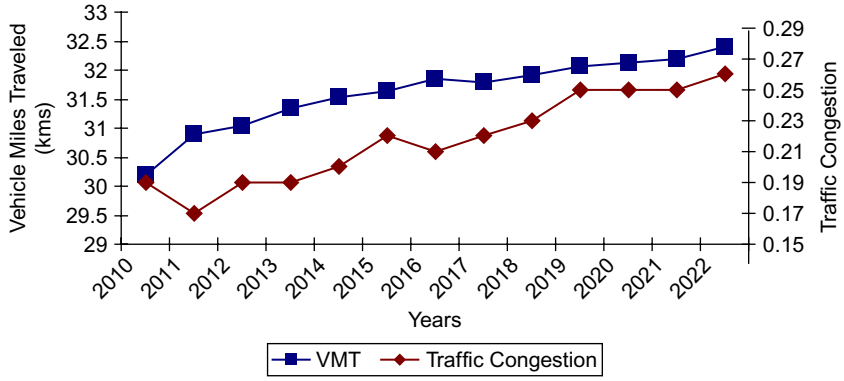


Figure 6.
Increase in VMT and traffic congestion in response to fuel efficiency improvements

variable ranging between 0 and 1 with congestion increasing from 0 to 1. In the simulation results (Figure 6), the traffic congestion is to vary in a range of 0.19-0.26, and the value of which can be even higher for Indian roads. The road infrastructure developments in the past years in India as captured in the model are mainly aimed for all the transport modes in the country. Hence the value for traffic congestion is low as only four-wheelers are considered in the present study and other modes of transport (like bus, three-wheelers, bicycles, pedestrians, etc.) making use of the same road space are not considered.

6.2.2 Effect of fuel efficiency improvements on direct rebound effect. From the system dynamic model developed, direct rebound effect (loop R1 in Figure 2) was measured using the identity presented in Section 3 based on price elasticity. The energy service cost (here travel cost) and demand for useful work (here four-wheeler distance traveled or VMT) were utilised for the same. The direct rebound effect graph in Figure 7 shows that its value fluctuates between -2 and 0 and is in accordance with the values obtained by Greene (2010) and Small and Van Dender (2007). For the present fuel efficiency improvement scenario with a *per capita* income growth rate of 1.7 percent, the estimated direct rebound effect for personal transport sector in India is found to increase over the time horizon. This in turn shows that the fuel efficiency improvement

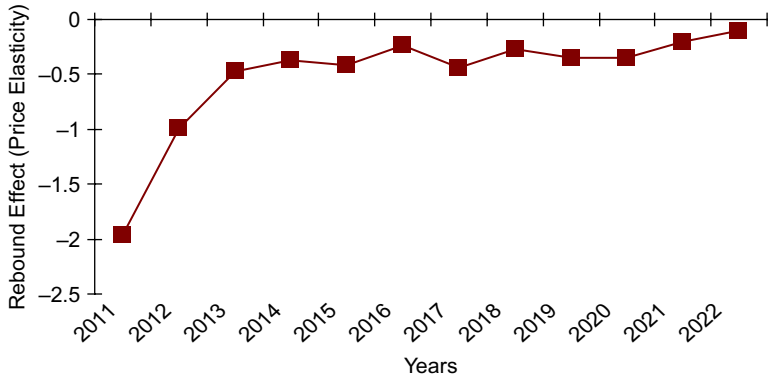


Figure 7.
Direct rebound effect in personal transport sector in India

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policy has a positive influence on direct rebound effect in personal transport sector in India and is in line with the observations of Roy (2000).

One of the current researches on rebound effect is on the matter of its stability over time (Greene, 2010). From Figure 7, it is evident that the rebound effect is not constant over time. It varies over time in proportion to the fuel cost share of the total travel cost. This result obtained through our simulation studies is in line with the observations of Gately (1992) and Small and Van Dender (2007).

6.2.3 Effect of fuel efficiency improvements on fuel consumption, imports and emissions. Fuel considered in the present study is gasoline and the price of gasoline is taken to be the total of international fuel price at which the fuel is imported and the fuel tax imposed by the government. This is because more than 85 percent of the fuel requirements in India are met through imports. In the present simulation, the "international fuel price" is considered as an exogenous variable. India imports transportation fuel as refined fuel as well as unrefined crude oil. In the present study, the "fuel imports" implies the refined fuel imports and unrefined crude oil imports together.

One of the motivations for fuel efficiency improvement made is to reduce the fuel consumption thereby leading to a reduction in imports to the country. From the simulation results (Figure 8) it is evident that the improvements in fuel efficiency of four-wheelers have not served this purpose. The fuel consumption and fuel imports have actually increased with time. This shows that in spite of the present fuel efficiency improvement scenario, India will become increasingly more dependent on fuel imports than it is now.

The increase in four-wheeler ownership and traffic congestion has further impacts on fuel consumption and emission levels in the country. Traffic congestion results in wastage of fuel thereby increasing air pollution in the form of carbon dioxide emissions. For the present study, The emissions are measured in metric tons of carbon dioxide equivalent emitted per year in India and is found to be of the order of 114,000 Mt from transport sector in the year 2010. The simulation result shows that the increase in fuel consumption owing to fuel efficiency improvements has resulted in an increase in emission levels in the country. With the present fuel efficiency improvement scenario, more than two times increase in fuel consumption and emission levels is expected within the next 12 years in the country (Figure 8).

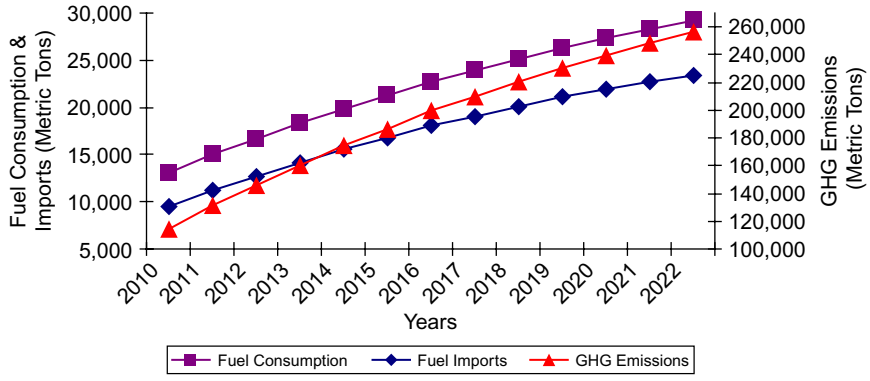


Figure 8.
Increase in fuel consumption, fuel imports and emissions in response to fuel efficiency improvements

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The simulation results for the important variables for scenario-1 and scenario-2 are summarized in Table III. Comparing the scenario-1 and scenario-2 from Table III, it is evident that the four-wheeler ownership, fuel consumption, fuel imports and emission are higher for scenario-2 than for scenario-1. This in turn clearly shows that the fuel efficiency improvement efforts have backfired through increasing the fuel consumption, imports and emissions in India.

7. Conclusions

In this study, we have contributed with:

- A system dynamics model.
- The assessment of the effects of fuel efficiency improvement policy on four-wheeler sector growth, transport sector fuel consumption, fuel imports, traffic congestion and emission in the context of India.

A causal loop model was proposed, which was transformed to a stock and flow simulation model based on the system dynamics methodology. Using the system dynamics simulation results, the conclusions made are:

- The fuel efficiency improvement policy will increase the demand for fuel efficient four-wheelers in the market thereby favouring the sector's growth.
- The growth in four-wheeler ownership, owing to the fuel efficiency improvements, increases the traffic congestion on Indian roads. This increased ownership and congestion levels in turn increases the fuel consumption, fuel imports and emissions in the country.
- Fuel efficiency improvements results in reduced travel cost which triggers more trips and hence the distance traveled resulting in a direct rebound effect. This effect adds up to the fuel consumption, fuel imports and emissions in the country.
- In the long-run the direct rebound effect is found to increase for the personal transport sector in India. Moreover, this effect was found unstable and varying over time.

Like in any other country, the policy of fuel efficiency improvements in Indian transport sector is mainly aimed at reducing the fuel consumption levels leading to reduced fuel imports and emissions. The conclusions of this study show that this policy has not served its intended purpose and instead has resulted in further increase in fuel consumption and related imports and emissions in the country. If the present fuel efficiency improvement scenario continues in future, we should expect more than two times increase in the levels of fuel consumption, fuel imports, and transport sector emissions in India. Hence in future, this policy should be implemented with care.

To reap the benefits of fuel efficiency improvements, other complimentary policies such as proportionate increase in fuel prices, car sharing, imposition of carbon tax, and increased road taxes (a policy that increases cost per mile), may be considered to assess their constraining effect on the vehicle distance traveled (VMT). The aim should be to create policy synergy that would bring about the best possible scenario in terms of fuel conservation and reduced emissions. The future work includes detailed study and dynamic analysis of the effects of these policies in the context of India.

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Appendix

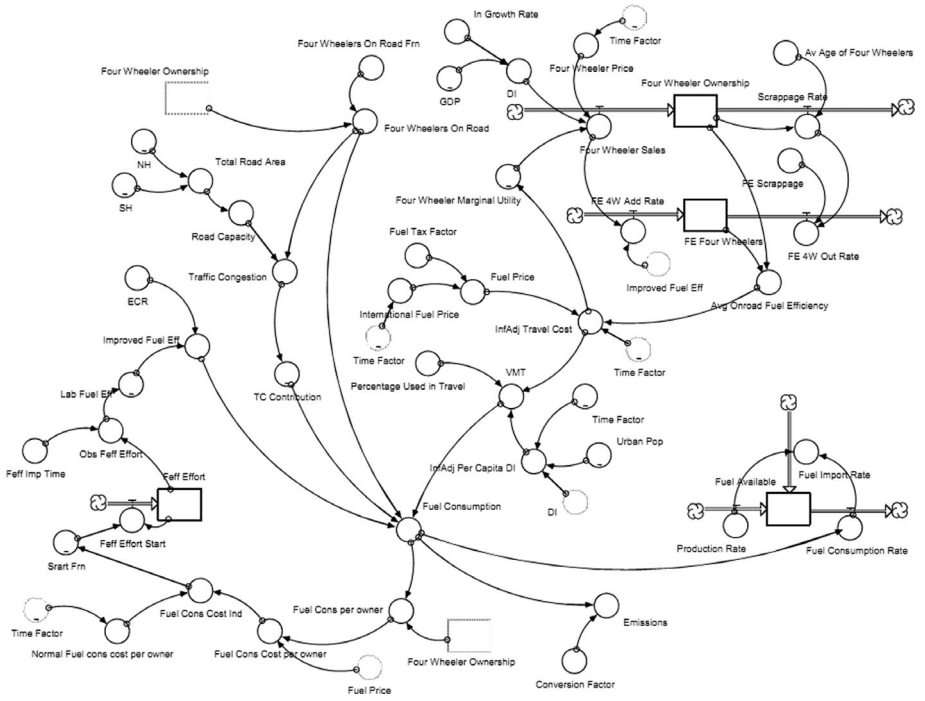


Figure A1.
Stock and flow diagram for the proposed causal loop model

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